

Assignment 7

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Problem 1

(a)

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a, b, c, d \in \mathbb{R}.$$

Compute

$$A^2 = \begin{pmatrix} a^2 + bc & b(a + d) \\ c(a + d) & d^2 + bc \end{pmatrix}.$$

The equation $A^2 = -I$ yields the system

$$a^2 + bc = -1, \quad b(a + d) = 0, \quad c(a + d) = 0, \quad d^2 + bc = -1.$$

If $a + d \neq 0$ then $b = c = 0$; but then $a^2 = -1$, impossible in $\mathbb{R}$. Hence $a + d = 0$, so $d = -a$. Substituting into $a^2 + bc = -1$ gives

$$bc = -(1 + a^2).$$

Thus $bc < -1$. Put $b = -p$, $c = -q$ so that $pq = 1 + a^2 > 1$. Then

$$a = \pm\sqrt{pq - 1}, \quad d = -a = \mp\sqrt{pq - 1}.$$

Hence every real 2×2 solution has the form

$$A = \begin{pmatrix} \pm\sqrt{pq - 1} & -p \\ -q & \mp\sqrt{pq - 1} \end{pmatrix}, \quad pq > 1,$$

where the two signs are chosen simultaneously (upper sign with lower sign opposite). Direct substitution verifies $A^2 = -I$.

(b)

Suppose $A \in \mathbb{R}^{2 \times 2}$ satisfies

$$A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 - \varepsilon \end{pmatrix}, \quad \varepsilon > 0.$$

Consider the eigenvalues of A . If A had two real eigenvalues λ_1, λ_2 , then A^2 would have eigenvalues $\lambda_1^2, \lambda_2^2 \geq 0$, contradicting that the right-hand side has strictly negative eigenvalues. If instead the eigenvalues of A are a nonreal complex conjugate pair $\alpha \pm i\beta$ ($\beta \neq 0$), then the eigenvalues of A^2 are $(\alpha \pm i\beta)^2$, which are conjugate to each other. Conjugate complex numbers cannot equal two distinct real numbers -1 and $-1 - \varepsilon$. Therefore no real 2×2 matrix A can satisfy the displayed equation when $\varepsilon > 0$.

Problem 2

Let

$$M = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

First diagonalize M . The characteristic polynomial is $(2 - \lambda)^2(1 - \lambda)$, so the eigenvalues are 2, 2, 1. One convenient choice of independent eigenvectors is

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Let $S = [v_1 \ v_2 \ v_3]$, i.e.

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

One checks that

$$S^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = S \operatorname{diag}(2, 2, 1) S^{-1}.$$

To find B with $B^4 = M$, take fourth roots of the diagonal entries (on the chosen eigenbasis). For the double eigenvalue 2 we may choose fourth roots of opposite signs to keep B real on that invariant subspace. Set

$$D = \operatorname{diag}(2^{1/4}, -2^{1/4}, 1).$$

Then define

$$\boxed{B = S D S^{-1}.$$

Because $D^4 = \operatorname{diag}(2, 2, 1)$, we have $B^4 = S D^4 S^{-1} = M$.

Expanding B gives the explicit real matrix (writing $r = 2^{1/4}$):

$$B = \begin{pmatrix} r & 0 & 0 \\ 0 & -r & 0 \\ 0 & r+1 & 1 \end{pmatrix}, \quad r = 2^{1/4}.$$

A direct check shows $B^4 = M$. (One way to verify is to compute SDS^{-1} symbolically or numerically and raise it to the fourth power; since B is similar to D , the relation $B^4 = M$ follows from similarity and $D^4 = \text{diag}(2, 2, 1)$.)

Problem 3

Let

$$N = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = E_{1,4}.$$

Note $N^2 = 0$ but $N \neq 0$. Thus N is nilpotent of index 2, has rank 1, and its Jordan canonical form consists of one 2×2 Jordan block and two 1×1 zero blocks; i.e. Jordan type $[2, 1, 1]$.

We ask for which integers n there exists a complex 4×4 matrix B with $B^n = N$.

Consider possible Jordan types for a nilpotent 4×4 matrix B : $[4]$, $[3, 1]$, $[2, 2]$, $[2, 1, 1]$, $[1, 1, 1, 1]$. The effect of powering on Jordan blocks is standard:

- If B has type $[4]$ (a single 4×4 Jordan block $J_4(0)$), then $J_4(0)^3$ is similar to a matrix with a single nonzero entry in the $(1, 4)$ position; in particular B^3 has Jordan type $[2, 1, 1]$. Hence $n = 3$ is possible (choose $B \sim J_4(0)$).
- If B has type $[3, 1]$, then B^2 acting on the 3×3 Jordan block yields a single Jordan block of size 2 together with zeros, so B^2 has type $[2, 1, 1]$. Thus $n = 2$ is possible.
- If B has type $[2, 1, 1]$, then B^1 itself already has type $[2, 1, 1]$; so $n = 1$ is possible.
- Type $[2, 2]$ cannot produce a rank-1 nilpotent for any positive power: powers either keep rank ≥ 2 (for small powers) or become 0 (for large powers).

Therefore the integers n for which such a B exists are

$$\boxed{n = 1, 2, 3.}$$

Problem 4

Let A be Hermitian, $A^H = A$. If $Av = \lambda v$ for $v \neq 0$, then

$$v^H Av = \lambda v^H v.$$

Taking conjugate gives

$$(v^H Av)^* = v^H A^H v = v^H Av = \bar{\lambda} v^H v.$$

Thus $\lambda v^H v = \bar{\lambda} v^H v$. Since $v^H v > 0$, it follows that $\lambda = \bar{\lambda}$, i.e. $\lambda \in \mathbb{R}$. Hence every eigenvalue of a Hermitian matrix is real.

Problem 5

Suppose A is Hermitian and satisfies

$$A^5 + A^3 + A = 3I.$$

Let $\lambda \in \mathbb{R}$ be any eigenvalue of A and v an associated eigenvector. Applying the equation to v yields the scalar equation

$$\lambda^5 + \lambda^3 + \lambda = 3.$$

Define $f(\lambda) = \lambda^5 + \lambda^3 + \lambda - 3$. Then

$$f'(\lambda) = 5\lambda^4 + 3\lambda^2 + 1 > 0 \quad \text{for all } \lambda \in \mathbb{R},$$

so f is strictly increasing on $\mathbb{R}$ and therefore has at most one real root. Since $\lambda = 1$ is a root, it is the unique real root. Thus every eigenvalue of A equals 1, so $A = I$.

Problem 6

Consider

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Compute $A - I$:

$$A - I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Perform elementary column/row reasoning to find $\text{rank}(A - I) = 2$. (For instance, the first column and the sixth column are independent and all other columns are linear combinations of them; one can check no single column is zero except the first row, and the two chosen are independent.) Therefore

$$\dim \ker(A - I) = 6 - \text{rank}(A - I) = 4.$$

Thus the geometric multiplicity of eigenvalue 1 is 4. The trace $\text{tr}(A) = 6$ implies the algebraic multiplicity of the eigenvalue 1 must be 6, so 1 is the only eigenvalue (with algebraic multiplicity 6).

Since there are 4 Jordan blocks (equal to the geometric multiplicity) whose sizes sum to 6, the only possible block-size partition is 2, 1, 1, 1, 1 (one 2×2 Jordan block and four 1×1 blocks).

We now present explicit vectors that realize a Jordan chain for the 2×2 block and some eigenvectors for the 1×1 blocks. Let

$$e_6 = (0, 0, 0, 0, 0, 1)^T, \quad v = (0, 1, 1, 1, 1, 1)^T.$$

A direct computation shows

$$(A - I)e_6 = 0, \quad (A - I)v = 4e_6.$$

Hence e_6 is an eigenvector and v is a generalized eigenvector with $(A - I)v$ proportional to e_6 . This provides a Jordan chain of length 2. The remaining four independent eigenvectors may be chosen from $\ker(A - I)$ to complete a basis of generalized eigenvectors.

Therefore the Jordan canonical form of A is

$$J = \text{diag}(J_2(1), 1, 1, 1, 1), \quad J_2(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Problem 7

Given $Ax = b$ and $(A + E)y = b$, assume $\|\cdot\|$ is an induced matrix norm and $\|A^{-1}E\| < 1$. We wish to bound $\|x - y\|/\|x\|$.

From $Ax = b$ and $(A + E)y = b$ we have

$$x - y = A^{-1}b - (A + E)^{-1}b = A^{-1}Ey.$$

Hence

$$\|x - y\| \leq \|A^{-1}\| \|E\| \|y\|.$$

From $(A + E)y = Ax$ we obtain

$$(I + A^{-1}E)y = x,$$

so

$$y = (I + A^{-1}E)^{-1}x.$$

Using the Neumann series and $\|A^{-1}E\| < 1$ we have

$$\|(I + A^{-1}E)^{-1}\| \leq \frac{1}{1 - \|A^{-1}E\|}.$$

Thus

$$\|y\| \leq \frac{1}{1 - \|A^{-1}E\|} \|x\|.$$

Combining with the previous estimate gives

$$\|x - y\| \leq \|A^{-1}\| \|E\| \frac{1}{1 - \|A^{-1}E\|} \|x\|.$$

Dividing both sides by $\|x\|$ and using $\kappa(A) = \|A\| \|A^{-1}\|$, we obtain

$$\frac{\|x - y\|}{\|x\|} \leq \kappa(A) \frac{\|E\|}{\|A\|} \cdot \frac{1}{1 - \|A^{-1}E\|}.$$